

VRANJE, Serbia

WMO: 134890

Lat: 42.550N

Lon: 21.917E

Elev: 434

StdP: 96.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-8.0	-13.5	1.2	-9.5	-10.9	1.5	-6.7	7.6	1.8	6.4	2.0	1.6	50	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.4	33.8	19.9	31.9	19.4	30.3	19.1	21.3	30.4	20.5	29.4	19.7	28.2	2.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.9	24.7	17.5	13.2	23.6	16.7	12.6	22.8	63.6	30.6	60.7	29.4	58.1	28.5	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.8	5.2	-14.2	36.4	3.4	1.8	-16.7	37.7	-18.7	38.7	-20.7	39.7	-23.2	41.0		
(5)			-14.6	22.8	3.3	1.2	-17.0	23.6	-19.0	24.3	-20.9	25.0	-23.3	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	0.3	2.7	6.8	11.8	16.1	19.8	22.1	22.4	17.6	12.1	6.9	1.6	(6)
(7)		DBStd	8.58	4.51	4.63	4.31	3.79	3.36	3.25	2.88	2.98	3.55	4.01	4.26	4.28	(7)
(8)		HDD10.0	1049	301	207	116	24	2	0	0	0	1	27	111	262	(8)
(9)		HDD18.3	2780	559	438	358	198	85	22	4	4	55	194	343	519	(9)
(10)		CDD10.0	1679	1	2	16	78	189	294	374	386	228	92	18	1	(10)
(11)		CDD18.3	368	0	0	0	1	14	66	121	132	33	2	0	0	(11)
(12)		CDH23.3	3958	0	0	1	28	176	696	1278	1395	354	31	0	0	(12)
(13)		CDH26.7	1506	0	0	0	2	24	229	529	614	104	2	0	0	(13)
(14)	Wind	WSAvg	2.1	2.1	2.3	2.5	2.3	2.1	2.2	2.0	1.9	1.8	1.9	2.1	(14)	
(15)	Precipitation	PrecAvg	613	42	40	41	58	62	56	47	41	55	63	55	52	(15)
(16)		PrecMax	932	104	95	99	179	128	136	114	125	162	137	131	102	(16)
(17)		PrecMin	344	7	4	4	15	13	5	4	2	13	4	1	1	(17)
(18)		PrecStd	140	26	20	23	35	28	30	30	32	33	38	36	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	17.1	21.3	26.5	29.2	33.4	35.9	36.5	32.6	26.5	20.1	14.4	(19)
(20)			MCWB	9.3	10.3	12.0	15.1	17.5	20.6	20.4	20.0	18.4	16.6	13.6	10.0	(20)
(21)		2%	DB	10.3	14.3	18.8	23.3	27.0	31.2	33.7	34.2	29.4	23.7	17.4	11.7	(21)
(22)			MCWB	7.3	8.8	10.7	14.0	16.8	20.1	19.9	19.8	17.6	15.2	12.5	9.2	(22)
(23)		5%	DB	8.5	11.9	16.6	20.9	25.4	29.4	31.7	32.1	27.2	21.4	15.4	9.8	(23)
(24)			MCWB	6.4	7.6	9.6	12.4	16.2	19.4	19.6	19.1	17.1	14.6	11.5	7.7	(24)
(25)		10%	DB	6.6	9.9	14.4	18.9	23.5	27.5	29.8	30.3	25.0	19.4	13.6	7.9	(25)
(26)			MCWB	5.0	6.7	8.5	11.6	15.4	18.6	19.2	18.7	16.4	13.8	10.5	6.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.1	13.0	16.2	18.8	22.7	22.1	22.1	20.1	17.5	14.4	11.1	(27)
(28)			MCD B	12.7	15.9	19.2	24.0	26.1	30.4	31.1	32.3	28.7	24.3	18.3	13.3	(28)
(29)		2%	WB	7.8	9.2	11.2	14.5	17.7	21.1	21.1	21.0	18.7	16.2	13.1	9.3	(29)
(30)			MCD B	9.8	13.1	17.3	21.6	25.0	29.0	30.6	30.7	26.9	21.8	16.4	11.4	(30)
(31)		5%	WB	6.5	8.0	10.1	13.3	16.9	20.0	20.4	20.1	17.8	15.2	12.0	7.9	(31)
(32)			MCD B	8.2	11.4	15.3	19.3	23.8	27.8	29.5	29.4	25.3	20.0	15.1	9.5	(32)
(33)		10%	WB	5.1	6.8	9.0	12.2	16.0	19.1	19.6	19.3	17.0	14.2	10.8	6.5	(33)
(34)			MCD B	6.6	9.8	13.9	17.7	22.3	26.3	28.2	28.2	23.4	18.5	13.3	7.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	8.4	9.8	10.8	11.4	12.0	12.9	13.4	11.7	10.8	8.5	6.1	(35)
(36)			MCDBR	9.3	12.8	14.7	14.9	14.9	15.5	16.5	17.0	15.9	14.8	11.3	8.6	(36)
(37)			MCWBR	6.5	7.8	7.7	6.8	6.3	6.0	5.6	5.5	6.3	7.5	6.8	6.2	(37)
(38)		5% WB	MCDBR	8.4	11.6	13.2	13.0	13.2	14.1	14.6	14.5	13.5	12.9	9.8	7.6	(38)
(39)			MCWBR	6.1	7.2	7.2	6.3	6.1	6.0	5.6	5.6	5.9	6.9	6.1	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.364	0.411	0.472	0.461	0.443	0.445	0.462	0.436	0.396	0.363	0.335	(40)	
(41)		taud	2.323	2.235	2.145	1.992	2.087	2.187	2.182	2.131	2.175	2.271	2.322	2.338	(41)	
(42)		Ebn at Noon	787	820	823	797	819	835	828	798	788	775	746	751	(42)	
(43)		Edh at Noon	86	110	135	170	159	144	143	146	130	105	86	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.54	2.43	3.46	4.46	5.42	6.23	6.46	5.80	4.18	2.94	1.86	1.31	(44)	
(45)		RadStd	0.24	0.32	0.42	0.55	0.45	0.53	0.47	0.48	0.34	0.38	0.20	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.69	N/A	N/A	+0.99	+0.55	+0.59	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		+0.54	N/A	N/A	N/A	+0.38	+0.39	N/A	-154	N/A	N/A

Nomenclature: See separate page